



POLICY

High Risk Medicines

Scope (Staff):	Medical, Dental, Nursing, Pharmacy, Anaesthetic Technicians
Scope (Area):	CAHS (CACH, CAMHS, PCH & Neonatology)

Child Safe Organisation Statement of Commitment

CAHS commits to being a child safe organisation by applying the National Principles for Child Safe Organisations. This is a commitment to a strong culture supported by robust policies and procedures to reduce the likelihood of harm to children and young people.

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Aim

To inform staff who prescribe, dispense, supply or administer medicines of the safe use of high-risk medicines within the Child and Adolescent Health Service (CAHS).

Risk

Inappropriate storage, handling, prescribing and administration of high-risk medicines increases the risk of harm and death to patients, non-compliance with legislation and Health Department policy, and increased cost to the organisation.

Background

All medicines carry risk of adverse events if prescribed, administered or dispensed incorrectly. High risk medicines are medicines that have a high risk of causing significant patient harm when they are used in error.^{1,2} Errors with these medicines may not be more common than those from other groups but their consequences can be more devastating.²

High risk medicines include medicines with a narrow therapeutic index and / or medicines that present a high risk of causing harm when administered by the wrong route or when other medication management system errors occur.² They are grouped nationally under the acronym 'APINCHS'. At a minimum CAHS high risk medicines include:

- A** Anti-infective agents
 - Aminophylline
 - Antiarrhythmics
- P** Potassium and other electrolytes
 - Positive Inotropes
 - Psychotropic agents
 - Phenobarbital (phenobarbitone)
 - Phenytoin
- I** Insulins
- N** Narcotics / opioids and sedatives
 - Neuromuscular blocking agents
- C** Chemotherapeutic agents and Immunomodulators
 - Captopril
- H** Heparin (including enoxaparin), other anticoagulants and thrombolytics

- S** Systems (e.g. independent second checks, safe administration of liquid medications, standardised order sets and medication charts etc.)

Principles

- CAHS is committed to ensuring that relevant high-risk medicine risk mitigation strategies (outlined below) are incorporated where relevant to all practices regarding storage, handling, prescribing and administration of these medicines.
- Clinical staff involved in the storage, handling, prescribing, dispensing and administration of high-risk medicines must be aware of the medicines on the CAHS High-Risk Medicines list (outlined below) and any other related policy documents.
 - The information presented in the CAHS High Risk Medicines List reflects what is currently regarded as safe practice but does not replace the need for the application of clinical judgement to each individual presentation.
- Every staff member has a responsibility to alert their manager if they identify any practice related to medication storage, handling, prescribing, dispensing or administration that places patient(s) at risk so that risk management strategies can be implemented to improve systems and processes.
- Before a new medicine is added to the Statewide Paediatric Medicines Formulary (Formulary One) it is reviewed by the CAHS Drug and Therapeutics Committee (DTC) and Western Australian Drug Evaluation Panel (WADEP) for safety, efficacy and cost effectiveness.
- Patients / carers must be included in the shared decision making process. A patient / carer has the right to a second opinion about any medication prescribed. Patients / carers must be provided with information regarding options for their care by the treating team, including the risks and benefits of medicines, and agreed goals of their treatment. This forms part of the consent process and must be documented in the patient's healthcare record.
 - For further information regarding Consent to Treatment refer to [CAHS Consent to Treatment Policy](#).
- If there is a high risk of death or serious injury to a patient where a medicine is inadvertently misused or incorrectly administered, the medicine is to be included in the High Risk Medicines Policy and an appropriate guideline developed.

High Risk Medicine Risk Mitigation Strategies ^{2,3,4}

A range of strategies should be considered to promote safer use or decrease the opportunity for misuse or error. The following are high risk medicine risk mitigation strategies:

Reducing or eliminating the risk of error

- Ensure the safest means of completing a task is efficient and easily followed.
- Remove opportunity for error by separating error-prone tasks from one another, in time and / or place.
- Set mechanical devices (i.e. smart infusion pumps) so that they default to the safest setting.
- Ensure the route of administration is clearly identified.
- If not administering by injection, use [oral / enteral syringes](#) to prepare and administer all liquid medicines.
- Remove or reduce the need for rapid mathematical calculation and reduce options and choices as far as possible through standardising concentrations of medicines in solution.
- Abide by the safety principles and recommendations of the Australian Commission on Safety and Quality in Healthcare Standards; including labelling requirements, visual product discrimination, reducing the amount of manipulation required to prepare a dose and associated equipment needed in administration.
- Ensure that strengths of medicines on labels, medication orders and on-screen displays etc. are clearly visible in terms of the dosage unit or dose per volume of liquid e.g. mg / mL.
- Limit access or use of high risk medicines.
- Maximise staff access to information about high risk medicines.
- Use Tallman lettering to differentiate high risk medicines.
- Use auxillary labels and automated alerts.
- Standardise the prescription, storage, preparation and administration of high risk medicines.
- Low level risk-reduction strategies (i.e. staff education and passive information) should be used together with high level risk reduction strategies such as force functions and fail-safes (e.g. limiting access or use, constraints, barriers or standardisation).
- Procure / purchase for safety as far as possible:
 - Review label clarity
 - Avoid look-alike and sound-alike medications
 - Avoid similar drug names
 - Only use patient's own medication after a prescriber or pharmacist has reviewed them for safety and approved their use
 - Ensure appropriate management of medication shortages

- Employ a fully [independent second check](#), carried out by a second authorised clinical staff member, for high risk medicines (as well as any medicine for which an independent second check is mandated by CAHS policy).
 - Refer to the [Medication Preparation, Checking and Administration Procedure](#) (PCH Medication Management Manual).
- Ensure all clinical staff have qualifications and skills appropriate to undertake prescribing, dispensing, supply, preparation and administration tasks required.
- Utilise bar-code scanning and medication automation technology to verify that the correct high risk medicine has been selected for the right patient (PCH only).

Making errors transparent

- Regularly review local and wider system medication incidents and near-misses.
- Use prospective analysis and system redesign to prevent recurrence of the same errors.

Minimising the consequences of error

- Stock low concentration and/or low volume of high risk medications so if inadvertently administered the risk of harm is minimised.

Monitoring patients receiving high risk medicines

- Patients receiving high risk medicines are vulnerable to medication error and must be closely and regularly monitored according to the relevant high risk medicine protocol.
- Where applicable, follow Therapeutic Drug Monitoring (TDM) recommendations for each high-risk medicine.
 - Results of drug levels and laboratory tests such as renal and liver function tests must be assessed to guide treatment.

Reviewing and learning for improvement

- All incidents regarding high-risk medicines must be reported via Datix Clinical Incident Monitoring System (CIMS) and reviewed in order to improve practice by implementing risk management or improvement strategies.
 - Refer to the [Clinical Incident Management](#) (CAHS Policy Manual).
- All available data sources such as clinical incident monitoring system data, internal reports and published articles will be regularly reviewed at the CAHS Medication Safety Committee meetings to identify any emerging or unidentified risks associated with the use of high risk medicines.

High Risk Medicines List

For specific drug administration advice, please refer to the [Medication Administration Policy](#) and individual [Medication Monographs](#).

Anti-infective Agents	
Refer to Children's Antimicrobial Management Program (ChAMP) policy.	
Drug / class and Resources	Description
<u>Aminoglycosides:</u> Amikacin Monograph Gentamicin Monograph Tobramycin Monograph Tobramycin (Inhaled) Monograph Gentamicin and Tobramycin Reporting Form (MR860.91)	Over dosing with respect to age, ideal body weight and renal function may result in significant ototoxicity and nephrotoxicity.^{4,5} Under dosing may result in treatment failure.⁴ Monitoring of serum levels, with appropriate dose adjustment, must be undertaken in all patients according to the ChAMP recommendations. Patients with unstable renal function must be monitored more frequently.^{4,5}
Liposomal amphotericin B (AmBisome®) Monograph Liposomal amphotericin B (Inhaled) Monograph (AmBisome®)	Confusion between the formulations of liposomal amphotericin B (AmBisome®) and amphotericin B (Fungizone®) may result in prescribing, dispensing or administration errors.⁴ Therefore, amphotericin prescriptions should state both the generic and trade names. Awareness of these multiple formulations and differing dosage recommendations will help reduce the risk of under- or overdosing.⁴
<u>Antifungal:</u> Itraconazole Monograph	Itraconazole is a broad spectrum azole antifungal. The two available capsule brands (Sporanox® and Lozanoc®) and oral liquid (Sporanox liquid®) are NOT interchangeable.⁶ Note: <i>Sporanox® capsules are not stocked at PCH</i> With the availability of three formulations from two different brands, there are potential prescribing, dispensing and administration risks.

Anti-infective Agents

Refer to [Children's Antimicrobial Management Program \(ChAMP\)](#) policy.

Drug / class and Resources	Description
	ALL prescriptions should state the formulation and brand required. One 50 mg Lozanoc[®] capsule is equivalent to one 100 mg Sporanox[®] capsule.⁶ The recommended dose for Lozanoc[®] is therefore half the recommended dose for conventional itraconazole capsules. Itraconazole has a number of clinically significant drug interactions that may affect drug levels and efficacy. Dose adjustment may be required.
<u>Antifungal:</u> Posaconazole Monograph	Posaconazole is a fungistatic azole that produces a higher incidence of neutropenia and liver toxicity than other azole antifungals.⁷ Oral bioavailability of the tablet formulation is four times that of the liquid.⁷ Changing from the liquid to tablet formulation can produce significant risk of increased toxicity if not intentional or without appropriate monitoring. Posaconazole has a number of clinically significant drug interactions that may affect drug levels and efficacy.⁷ Dose adjustment may be required.
<u>Antifungal:</u> Voriconazole Monograph	Voriconazole has non-linear pharmacokinetics and any dose adjustments should be made with caution.⁸ Inter and within-patient variability is significant and therefore therapeutic drug monitoring is essential.⁸ <i>Bioequivalence between the oral tablets and suspension has <u>not</u> been demonstrated. Higher doses are likely to be required for patients < 15 years and < 50 kg.⁸</i> Voriconazole has a number of clinically significant drug interactions that may affect drug levels and efficacy.⁸ Dose adjustment may be required.

Anti-infective Agents

Refer to [Children's Antimicrobial Management Program \(ChAMP\)](#) policy.

Drug / class and Resources	Description
<u>Antivirals:</u> Aciclovir Monograph Cidofovir Monograph Ganciclovir Monograph Valaciclovir Monograph Valganciclovir Monograph	Aciclovir, cidofovir, ganciclovir, valaciclovir and valganciclovir are different antivirals available as different formulations.^{9,10} They are renally cleared medicines and hence require maintenance of adequate hydration during therapy with consideration for dose modification in renal impairment.^{9,10} The names of these agents sound and look similar; confusion may lead to prescribing, dispensing or administering the wrong agent. Neurological effects including confusion, hallucinations, agitation, seizures and coma occur in over-dose. Cidofovir and ganciclovir must always be prepared by Pharmacy Compounding Services (PCS). They are potentially mutagenic, teratogenic and carcinogenic, and should be treated as cytotoxic.^{9,10}
<u>Antibacterial:</u> Colistimethate sodium Monograph Colistimethate sodium (inhaled) Monograph	Colistimethate sodium is a polymyxin antibacterial used to treat multi drug resistant gram negative bacterial infections.¹¹ It is available as an intravenous formulation or an inhaled formulation.¹¹ Colistimethate sodium and colistin base are not interchangeable. Doses are expressed as colistin base (mg) for intravenous therapy and colistimethate sodium (international units) for inhaled therapy.¹¹ Doses should be calculated on ideal body weight.¹¹ Colistimethate sodium is nephrotoxic - ensure renal function is monitored during systemic treatment with consideration for dose modification in renal impairment.¹¹

Anti-infective Agents

Refer to [Children's Antimicrobial Management Program \(ChAMP\)](#) policy.

Drug / class and Resources	Description
<u>Glycopeptides:</u> Teicoplanin Monograph Vancomycin (Intravenous) Monograph Vancomycin (Inhaled) Monograph Vancomycin (Oral) Monograph	Over dosing with respect to weight and renal function may cause nephrotoxicity and ototoxicity.⁴ Under dosing may result in treatment failure.⁴ Monitoring of serum levels, with appropriate dose adjustment, must be undertaken in all patients⁴ according to the ChAMP recommendations. Patients with unstable renal function and those on other nephrotoxic agents must be monitored more frequently.⁴
<u>Antibacterial:</u> Rifampicin Monograph	Rifampicin has been shown to cause liver dysfunction.¹² There have been fatalities associated with jaundice in patients with existing liver disease prescribed rifampicin and in patients receiving rifampicin concomitantly with other hepatotoxic agents.¹² There are a number of clinically significant drug interactions as rifampicin is a potent inducer of cytochrome P450 enzymes.¹² Dose adjustments may be required.¹²

Aminophylline

Drug / class and Resources	Description
Aminophylline Monograph	Aminophylline has a narrow therapeutic range and requires careful administration and monitoring as toxicity can occur even at therapeutic levels.¹³ Intravenous aminophylline use on general wards must be authorised by a Consultant / Fellow / Registrar of the Respiratory or General Paediatric Team.

Aminophylline	
Drug / class and Resources	Description
	A medical officer must review patients prior to each dose of intermittent aminophylline infusion.

Antiarrhythmics	
Drug / class and Resources	Description
<u>Antiarrhythmics</u> Amiodarone Monograph Digoxin Monograph Flecainide Monograph Isoprenaline Monograph Verapamil	Antiarrhythmic medications are associated with potential for serious adverse side effects, including proarrhythmic potential and a number of potentially serious drug interactions. Careful monitoring is required, particularly with use of IV antiarrhythmics. The following restrictions apply: • IV / Oral amiodarone – for prescription under the direction of a Cardiologist or Neonatologist • IV / Oral digoxin – for prescription under the direction of a Cardiologist • IV / Oral flecainide – for prescription under the direction of a Cardiologist. A Cardiologist must be present if IV flecainide is administered to a neonate at KEMH. At PCH a Cardiologist, Anaesthetist, Neonatal Intensive Care or Paediatric Critical Care Consultant must be present. • IV verapamil - for prescription by a Cardiologist or Nephrologist. • Oral verapamil – for prescription under the direction of a Cardiologist or Nephrologist. Care should be taken during the prescribing, dispensing and administration of antiarrhythmics to ensure the correct route and calculation of the rate of infusion is prescribed and administered to the patient.

Antiarrhythmics	
Drug / class and Resources	Description
	Continuous monitoring of blood pressure, ECG and heart rate must occur during intravenous administration.^{14,15} Amiodarone can cause or further decrease bradycardia and hypotension during intravenous infusion.^{14,15} Oral dosing often requires higher frequency (i.e. three times daily or twice daily dosing) when initiated which poses a long term risk of toxicity if not reduced appropriately.^{14,15} Always ensure appropriate monitoring and follow up. Digoxin dosing requires close monitoring and adjustment throughout the course of treatment.¹⁶ Dosing often requires higher frequency (i.e. three times daily or twice daily) when initiated, which poses a long term risk of toxicity if not reduced appropriately. Similarly, digoxin levels can be increased in the body by other medications or reduced renal function.¹⁶ For some indications, therapeutic drug monitoring is required.¹⁶ Digoxin oral paediatric liquid has a concentration of 50 micrograms/mL; doses to be administered are often less than a 1 mL volume. Careful counselling to consumers is important to prevent inadvertent overdosing.

Potassium and Other Electrolytes	
Drug / class and Resources	Description
Potassium Chloride Monograph Mandatory Standard for Intravenous Potassium (WA Health) Potassium Poster	Potassium salts are administered intravenously to replace clinical deficiency in patients who cannot receive the electrolyte orally or when urgent replacement is required.¹⁷ Errors in intravenous potassium preparation and administration can be fatal.¹⁷ Errors may be due to:

Potassium and Other Electrolytes	
Drug / class and Resources	Description
	Too rapid administration - Patients can suffer fatal cardiac events as a result of receiving a potassium chloride infusion too rapidly, even when appropriately diluted.¹⁷ Selection of the wrong ampoule – Potassium chloride ampoules can be mistaken for ampoules of similar appearance, for example, sodium chloride 0.9% when reconstituting a drug for injection. Consequently, the patient can be administered an unintended bolus of potassium.¹⁷ Preparation error - An intravenous infusion of potassium chloride prepared incorrectly, for example, not diluted appropriately (mixed with volume less than that required for the dose prescribed), not well mixed to distribute the potassium salt throughout the infusion fluid or potassium added to the infusion bag once it has been hung for administration. Extra potassium chloride MUST NEVER be added to standard (pre-mixed) solutions or compounded bags that already contain potassium as this may lead to confusion regarding the final concentration.¹⁷ When potassium chloride is added to an intravenous fluid, the solution must be inverted at least TEN times to ensure that the potassium chloride is thoroughly mixed throughout the solution.¹⁷ Use of an excessively strong solution¹⁷ - Patients can suffer fatal cardiac events as a result of receiving concentrated potassium chloride as a direct push injection or when potassium chloride concentrate has been added to an infusion without mixing the fluid bag prior to administration (see Preparation error). **Note: <i>the above information relates to the preparation of potassium chloride, potassium acetate and potassium dihydrogen phosphate.</i> Potassium chloride ampoules are kept in pharmacy and restricted clinical areas where they are used at a higher infusion rate or as a more concentrated solution. Use standard (premixed) potassium chloride intravenous solutions from imprest wherever possible.¹⁷

Potassium and Other Electrolytes

Drug / class and Resources	Description
	Solutions with a concentration greater than 40 mmol/L require administration via a Central Venous Access Device (CVAD). In exceptional circumstances, concentrations up to 80 mmol/L may be administered peripherally ONLY in Critical Care areas (Paediatric Critical Care and Emergency Department) at the discretion of the treating Consultant. Any patient prescribed potassium at a rate exceeding 0.2 mmol/kg/hour must receive continuous cardiac monitoring.
Other electrolytes: Calcium Monograph Magnesium Monograph Phosphate Monograph Sodium Bicarbonate Monograph Sodium Chloride 3% and 23.4% (Hypertonic Saline) Monograph	Concentrated electrolytes including calcium, magnesium, phosphate, sodium and bicarbonate pose a risk to patients when given in error. Multiple salts may be available. Check product selection carefully.

Positive Inotropes

Drug / class and Resources	Description
Positive Inotropes: Adrenaline (epinephrine) Monograph	Vasoactive drugs are utilised in the critical care setting for management of patients with cardiovascular and / or respiratory instability. Failure to manage these drugs appropriately can lead to haemodynamic instability and in extreme cases, death.

Positive Inotropes	
Drug / class and Resources	Description
Dobutamine Monograph Dopamine Monograph Isoprenaline Monograph Levosimendan Monograph Metaraminol Monograph Milrinone Monograph Noradrenaline (norepinephrine) Monograph	Inotropes are restricted for use in emergency situations and infused in critical care areas only. Milrinone has both inotropic and vasodilator actions and can cause hypotension, somnolence and arrhythmias in more than 1% of patients.¹⁸

Psychotropic agents	
Drug / class and Resources	Description
Clozapine Monograph Psychotropic medication - monitoring adverse physical health effects policy. Psychotropic Medication: Physical Health Monitoring Handbook CAMHS Guidelines for the use of the WA Clozapine	Clozapine requires strict monitoring in light of its potential to cause neutropenia, agranulocytosis, myocarditis, cardiomyopathy and clozapine induced gastrointestinal hypomotility (CIGH).⁴ Other significant adverse effects include marked sedation, weight gain, dyslipidaemia and hyperglycaemia, lowered seizure threshold, hypersalivation, asthenia, and sedation.⁴ Its prescription is limited to registered centres and all patients must be part of a clozapine monitoring system which mandates systematic evaluation of haematological parameters.⁴

Psychotropic agents	
Drug / class and Resources	Description
Initiation and Titration Chart (WA Health).	Concomitant use of other medications with the potential to cause agranulocytosis is discouraged and increased vigilance must be maintained where this is necessary.⁴ Tobacco smoking and caffeine alter the clearance of clozapine; dose adjustments must be considered and carefully monitored if intake is varied.¹⁹ The WA Clozapine Initiation and Titration chart must be used for all new patients commenced or restarted on clozapine.⁴
Lithium Monograph	There are relatively narrow margins between therapeutic and toxic dosages for lithium.⁴ Failure to recognise the early signs of toxicity may lead to a delay in treatment and result in poor patient outcomes including death.⁴ Toxicity usually occurs at concentrations greater than 1.5 mmol/L, but may develop at considerably lower concentrations.⁴ It is important to frequently monitor patient parameters including renal function, thyroid function, urea and electrolytes and lithium levels to ensure toxicity does not eventuate.⁴ The most significant contributors to lithium toxicity are⁴: • interactions with drugs that affect renal function (e.g. diuretics, nonsteroidal anti-inflammatory drugs (NSAIDs), angiotensin converting enzyme (ACE) inhibitors or angiotensin II receptor blockers) • reduced fluid or salt intake • fluid loss from vomiting, diarrhoea or excessive sweating. Toxicity may also be caused by deliberate or inadvertent overdose. Lithium is available in 250 mg immediate release tablets and 450 mg (scored) controlled release tablets. All medication orders should clearly state the formulation as the pharmacokinetics of these products are vastly different.

Psychotropic agents	
Drug / class and Resources	Description
<u>Parenteral antipsychotics:</u> Droperidol Monograph (except when used for nausea and vomiting) Haloperidol Zuclopenthixol Monograph	Avoid unnecessary IM or IV administration of antipsychotics choosing the oral route wherever possible. Ensure the child is monitored for sedation, QT prolongation and arrhythmias. In 2001 the FDA in the USA issued a Black Box warning as QT prolongation and/or torsades de pointes was reported in patients receiving droperidol at doses at or below recommended doses. Some cases have occurred in patients with no known risk factors for QT prolongation and some cases were fatal.²⁰ Children and adolescents are at greater risk for acute dystonic reactions than adults; close monitoring is required.²¹ Haloperidol has been associated with tardive and withdrawal dyskineasias in children, which may not be reversible.²¹ Zuclopenthixol acetate <u>must</u> only be prescribed under the direction of Psychiatrist. It poses a significant risk of causing extrapyramidal side effects and QTc prolongation.²² Due to its long half life the effects may last up to 72 hours.²³

Phenobarbital	
Drug / class and Resources	Description
Phenobarbital (phenobarbitone) Monograph	Rapid administration of IV phenobarbital may cause respiratory and circulatory collapse.²⁴ Resuscitation and artificial ventilation equipment must be readily available.²⁴

Phenytoin	
Drug / class and Resources	Description
Phenytoin Monograph	Phenytoin has a narrow therapeutic index. A small change in dose may result in a disproportionately large change in phenytoin concentration due to non-linear pharmacokinetics and saturation of it's hepatic metabolism.²⁵ Similarly a few missed doses or small change in drug absorption may result in a marked change in phenytoin concentration.²⁵ Preparations containing phenytoin sodium are not bioequivalent to those containing phenytoin base. Concentration monitoring is recommended when making changes between formulations.²⁵ Rapid administration of IV phenytoin may cause severe hypotension and cardiac arrhythmias (these events have also been reported at or below the recommended infusion rate).¹⁹ Blood pressure, ECG and respiratory function monitoring is recommended during IV use.^{25,26}

Insulin	
Drug / class and Resources	Description
Insulin Monograph Diabetic Ketoacidosis – Assessment and Management Insulin Poster	Errors in insulin therapy can cause serious harm, coma or death.⁴ Selecting the wrong product for administration may occur because of sound-a-like product names. The administration of an intravenous glucose 5% infusion must be run concurrently with any intravenous insulin infusion to prevent sudden severe hypoglycaemia. The following actions are to be used to promote safe and appropriate doses of insulin: • selection of the appropriate medication chart

Insulin	
Drug / class and Resources	Description
	• prescription to include full trade name e.g. Humulin NPH® not Humulin® • sensible dose rounding and avoidance of decimal points where possible. • administration device e.g. ○ intravenous insulin must be administered using a smart infusion pump ○ intermittent subcutaneous insulin must be administered using a pen device. Most pen devices measure insulin in whole units but PCH also stocks devices that allow dosing in half units. • time of administration (e.g. specific in respect to food) • dose written with the word 'UNITS' (except when the word 'UNITS' is already pre-printed on the chart). Clinicians must be competent in managing insulin therapy prior to prescribing or administering doses of insulin. All insulin prescriptions are to be written by specific brand name and strength required to eliminate confusion and error of using the wrong type and strength of insulin required.⁴

Narcotics (Opioids) and Sedative Agents	
Drug / class and Resources	Drug / class and Resources
<u>Opioids / Sedatives:</u> Alfentanil Dexmedetomidine Monograph Fentanyl Monograph Hydromorphone Monograph	These medicines have the potential for addiction and death if they are used inappropriately. Prescription Overdosing of opioids and sedatives can lead to excessive sedation and potentially lethal respiratory depression.⁴ Dosing should follow the “start low and go slow” philosophy.⁴ Prescribing opioids in patients with impaired renal clearance

Narcotics (Opioids) and Sedative Agents

Ketamine Monograph Morphine Monograph Opioids, benzodiazepines and other Schedule 8 medicines Oxycodone Monograph Pethidine (Intravesical) - Monograph Remifentanyl Tramadol	and in overweight or obese patients requires careful dose consideration and monitoring. Dosages should always be carefully titrated. When changing between two opioids it is recommended to refer to an opioid conversion guide or appropriate reference e.g. AMH CDC, and to carefully consider any breakthrough doses that have been used when calculating equianalgesic doses. Prescribers should contact the Acute Pain Service for advice. Patients may vary in their response to the effects of different opioids, therefore ongoing patient assessment is required after conversion. The risk of inadvertent overdose, over sedation or respiratory depression from concurrent prescription of other opioid or sedative medications must be considered, including patients who have received intrathecal or epidural opioids peri-operatively.⁴ Administration Cumulative sedation from medications used for symptomatic relief post surgery can occur when doses are co-administered or administered too frequently e.g. prescribing and administering benzodiazepines for muscle spasm or opioids second hourly for pain.⁴ Oral opioids should not be given while a patient is receiving opioids via patient / nurse controlled analgesia or continuous infusion post surgery. For all infusions, nurses must independently second check that the concentration and the infusion rate is consistent with the prescription.⁴ Regularly review the patient using the appropriate pain scoring tool to assess analgesia. Monitor sedation levels.⁴
<u>Opioid Transdermal Patch Delivery Systems:</u> Fentanyl Monograph	Fentanyl is a potent opioid and must only be prescribed under the direction of the Acute Pain Service, Palliative Care Physician, Haematologist, Oncologist or an Anaesthetist. Fentanyl is available as a patch formulation. Absorption of fentanyl from a patch can be increased by exposing the patch to heat sources such as heat packs, electric blankets and hot baths. A high fever can also increase absorption.⁴

Narcotics (Opioids) and Sedative Agents

	Ensure that the previous patch has been removed before applying a new patch.⁴ Both the application and removal of patch must be recorded on the medication chart.⁴ Ensure that the medicated patch has been removed if the prescription order has been ceased.⁴ Never use in opioid-naïve patients and only for chronic pain management.⁴ A significant amount of the drug remains in the patch after its intended application period has expired.⁴ The patch should be folded in half so that the sticky side of the patch sticks to itself and discarded into a secure sharps or medication bin.⁴
<u>Opioid Oromucosal Formulations:</u> Fentanyl Monograph	Fentanyl is a potent opioid and must only be prescribed under the direction of the Acute Pain Service, Palliative Care Physician, Haematologist, Oncologist or an Anaesthetist. Fentanyl lozenges should not be used in opioid-naïve patients because of the risk of respiratory depression. Prevent accidental or deliberate misuse by correct use, storage and disposal of fentanyl lozenges. Fentanyl lozenges should not be chewed or swallowed.²⁷ There is no dose equivalence between fentanyl lozenges and other opioid formulations. The optimal dose cannot be predicted by the dose of regular opioid or previous breakthrough opioid. It should be individually titrated by starting at the lowest dose.²⁷
<u>Benzodiazepines:</u> Clobazam Monograph Clonazepam Monograph Clonazepam Poster Diazepam Monograph Lorazepam	Overdosing of benzodiazepines can lead to excessive sedation and potentially lethal respiratory depression.⁴ Benzodiazepines should be prescribed according to ideal body weight in overweight or obese patients.²⁸ Dosages should always be carefully titrated. The risk of inadvertent overdose, over sedation or respiratory depression from concurrent prescription of opioid or sedative medications must be considered.⁴ Use of benzodiazepines can lead to physical and psychological dependence (even if used for short periods), tolerance and misuse.²⁹

Narcotics (Opioids) and Sedative Agents

Midazolam Monograph Nitrazepam Temazepam	Avoid concurrent use of IM benzodiazepines and IM olanzapine. Concurrent use may cause cardiorespiratory depression and excessive sedation. If the use of both is required they must be separated by at least 1 hour.³⁰ Clobazam and clonazepam look and sound similar but are different drugs with different indications and dosage. Confusion with these agents may lead to the prescribing, dispensing or administration of the incorrect medication. Patients may also be confused when requesting repeat outpatient prescriptions, increasing the risk of error during prescribing and dispensing. The clonazepam dose must always be prescribed as the number of drops and milligrams to prevent medication errors.
Chloral hydrate	Chloral hydrate is used off-label for painless procedural sedation and is most successful for patients less than 2 years of age and under 15 kg.³¹ Ensure adequate monitoring for respiratory depression and bradycardia.³¹ Drug accumulation and toxicity have occurred with repeated dosing.³¹
Clonidine Monograph	Clonidine has a narrow therapeutic index.³² When administered intravenously, clonidine is highly sedating. Overdose can cause life-threatening central nervous system (CNS) and respiratory depression.³² Avoid ceasing clonidine abruptly as it may precipitate a severe withdrawal syndrome.³³
Propofol Monograph	Propofol is used to induce anaesthesia and for procedural sedation amongst other uses.³⁴ Propofol infusion syndrome, involving arrhythmias, progressive myocardial failure, renal failure, metabolic acidosis, rhabdomyolysis, fatty liver and death has occurred. The risk appears to be greater with infusions > 48 hours and high infusion rates.³⁴ Propofol must only be used by trained personnel with advanced airway management skills i.e. Anaesthesia staff, staff in the Emergency Department and Paediatric Critical Care. Propofol must be administered under direct supervision of medical staff. Use outside of theatre and critical care

Narcotics (Opioids) and Sedative Agents

settings must be under the direct supervision of an Anaesthetist.

Neuromuscular blockers

Drug / class and Resources

Drug / class and Resources

Neuromuscular blockers:

[Atracurium Monograph](#)

[Rocuronium Monograph](#)

[Suxamethonium Monograph](#)

[Vecuronium Monograph](#)

Neuromuscular blocking drugs produce skeletal and respiratory muscle relaxation, during general anaesthesia. They have no sedative or analgesic effects.

Selection errors can have fatal consequences. These agents are to be used in emergency and anaesthetic circumstances only by or under the direct supervision of an anaesthetist / doctor skilled in management of artificial respiration and only when facilities are instantly available for endotracheal intubation and providing adequate ventilation of the patient.⁴

It is recommended that an auxillary label be adhered to all storage containers stating “ WARNING: Paralysing agent – causes respiratory arrest – patient must be ventilated”.⁴

Chemotherapeutic Agents and Immunomodulators

Drug / class and Resources

Description

Chemotherapeutic Agents and Immunomodulators

[See Individual Drug Monographs](#)

[Antineoplastic \(cytotoxic\) Agents Safe Handling and](#)

All chemotherapeutic agents are considered high-risk medications.⁴

Staff must follow local procedures and policies to ensure the safe preparation, administration and monitoring of chemotherapeutic agents including fulfilling the required competencies prior to prescription or administration of doses.

Chemotherapeutic Agents and Immunomodulators	
Drug / class and Resources	Description
Administration (PCH Medication Management Manual)	For local therapeutic information about other chemotherapeutic and immunomodulatory medicines not listed below contact Oncology Pharmacy directly.
Azathioprine and 6-mercaptopurine	Immunomodulatory thiopurines require close monitoring due to the risk of serious, potentially life-threatening adverse effects related to their metabolites.⁴ Myelosuppression and liver toxicity from thiopurines can occur anytime during their use and can be sudden in onset.⁴ Azathioprine and 6-mercaptopurine have a number of clinically significant drug interactions that may cause severe myelosuppression.⁴ There is potential for mercaptopurine to be confused with mercaptamine (cysteamine) due to names that look and sound alike. This may lead to prescribing, dispensing or administering the wrong agent.
Bleomycin	Doses of bleomycin must be expressed in International Units not in milligrams, 'mg', Units, 'U' or 'IU'. 1500 international units of bleomycin are equivalent to 1.5 USP units and approximately equivalent to 1.5 mg (by potency) or 1 mg (by weight).³⁵ Caution is required when converting from mg to international units as protocols or trials may state bleomycin in terms of mg-potency rather than mg-weight.³⁵ A test dose of bleomycin prior to administration of a dose is not used by all centres as the usefulness for predicting hypersensitivity reactions has been questioned.³⁵
Dinutuximab Monograph Dinutuximab BETA Monograph	Dinutuximab should not be confused with dinutuximab beta; these two monoclonal antibodies are not interchangeable.
Etoposide Monograph	Etoposide is available as the base drug etoposide and as the phosphate salt.⁴ Etoposide phosphate is a pro-drug and is not directly interchangeable with etoposide.⁴

Chemotherapeutic Agents and Immunomodulators	
Drug / class and Resources	Description
Etoposide PHOSPHE Monograph	Confusion when prescribing or administering the medication can result in under or over dosing. ⁴
Methotrexate	Australian cases with fatal consequences have been reported when oral methotrexate has been prescribed and administered more frequently than once weekly for autoimmune or inflammatory disorders.⁴ A monitoring and reversal protocol must be available when using high dose methotrexate (HD-MTX) regimens (> 1 g/m²). High morbidity and mortality have been reported in patients receiving these regimens and ONLY experienced and qualified staff can manage a patient receiving high dose methotrexate.
Rituximab Monograph - Paediatric Specialty Infusion Charts	Rituximab has complex monitoring requirements and has been included to ensure monitoring is carried out by staff prior to administration. The Rituximab VARIABLE RATE infusion chart should be used for all infusions.
<u>Vinca alkaloids:</u> Mandatory Standard for Vinca Alkaloids (WA Health) Vinblastine Monograph Vincristine Monograph Vinorelbine Monograph	Vinca alkaloids can be fatal if given by the intrathecal route.⁴ Vinca alkaloids are never to be stored in the same location as intrathecal products. All vinca alkaloids must be made up in a minibag. A warning label must be attached to the IV minibag and outer bag in keeping with the Product Information, this warning should read: "WARNING - For intravenous use only. FATAL if given by any other route".³⁶

Captopril	
Drug / class and Resources	Description
Captopril Monograph	Neonates and young infants are more sensitive to adverse effects and a test dose is recommended.³⁷ Overdose can be fatal – associated with significant seizures, apnoea and renal complications.³⁷ The oral liquid formulation has a concentration of 5000 microgram/mL (5 mg/mL). For all doses less than 1000 microgram (1 mg) the captopril 5000 microgram/mL (5 mg/mL) oral solution <u>must</u> first be diluted to allow an accurate dose to be measured prior to administration. Use extra caution to prevent inadvertent overdosing.

Heparin, other anticoagulants and thrombolytics	
There is potential for excessive bleeding with warfarin, heparin, other anticoagulants and thrombolytics. The incorrect dose or failure to monitor therapy can contribute to this event. Planning in advance for surgery and other procedures is required.	
Drug / class and Resources	Description
Enoxaparin Monograph Heparin Monograph Taurolidine / Sodium citrate / Heparin 100 units (TaurolockHep100®) Monograph	Heparins can be sub-classified as unfractionated heparin (UFH) and the low molecular weight heparins (LMWH) such as enoxaparin.⁴ These are available in different concentrations. Exercise caution when prescribing, dispensing and administering heparins. There is an increased risk of error with heparin because it is used for numerous indications with different dosing regimens. Ensure that the word 'UNITS' is written in full for each heparin prescription to avoid confusion and unintentional 10x overdoses.⁴ Store heparin according to local guidelines to prevent selection of an incorrect strength heparin ampoule. It is recommended that prescribers contact the Haematology team for advice for all paediatric patients receiving heparin.

Heparin, other anticoagulants and thrombolytics

There is potential for excessive bleeding with warfarin, heparin, other anticoagulants and thrombolytics. The incorrect dose or failure to monitor therapy can contribute to this event. Planning in advance for surgery and other procedures is required.

Drug / class and Resources	Description
Warfarin Monograph	Warfarin requires frequent monitoring and has the potential to interact with many medications that can affect the patient's International Normalised Ratio (INR).⁴ Two brands of warfarin (Marevan® and Coumadin®). are currently marketed in Australia. They are not bioequivalent and have different tablet strengths available.⁴ Be aware that changing diet, concurrent medications (by ceasing or adding a medication) or changing brand of warfarin may result in changes to the INR and the dose of warfarin may require adjustment.⁴
<u>Direct Acting Oral Anticoagulants (DOACs):</u> Rivaroxaban	Rivaroxaban has no specific reversal agent.⁴ Care is required when selecting patients for newer anticoagulant treatment and the following must be considered⁴: • Dose adjustment is required in renal impairment. • Check for drug interactions and use of other anticoagulants, antiplatelet agents and non steroidal anti-inflammatory drugs (NSAIDs). • Check for any existing co-morbidities with high risk of bleeding. A Haematologist must be consulted prior to use.
<u>Thrombolytics:</u> Alteplase Monograph	Alteplase is associated with a potential for serious adverse effects including intracerebral haemorrhage or other serious bleeding and allergic reactions.³⁸ Patients must be closely monitored. A Haematologist must be consulted prior to use of alteplase for systemic thrombolysis.

Systems	
System and Resources	Description
Independent second check⁴ Medication Administration Policy Medication Preparation, Checking and Administration Procedure Take 5 - Independent Second Check Process	Only certain medications can be single checked; all remaining medications must have an independent second check.
Safe Administration of Oral / Enteral Solutions Medication Administration Policy WA Health Medication Safety Resources - Oral Liquid and Oral Syringes	All oral / enteral solutions are to be administered using specific purple oral / enteral syringes that are distinguishable from syringes used for any other purpose.⁴
Smart Infusion Pumps Smart Infusion Pumps and DERS Policy Smart Infusion Pumps My Information Hub	Smart Infusion Pumps allow for the safe prescribing and administration of intravenous medications through the utilisation of Dose Error Reduction Software (DERS). DERS is a system of pre-programmed drug libraries containing medication dosing guidelines specific to PCH. In the interests of patient safety and safe administration practice, PCH Drug Libraries must be used at all times. * An exception to this is in an emergency at the discretion of the treating doctor for the duration of the emergency. The infusion should be reprogrammed using DERS as soon as possible.

Systems	
System and Resources	Description
Standardisation in Paediatric Prescribing WA HMC User guide Australian Commission on Safety and Quality in Health Care position statement on paediatric prescribing	The WA paediatric Hospital Medication Chart (pHMC) must be used for all hospital patients and patients receiving treatment at Specialised CAMHS sites. All anti-cancer therapy orders must be prescribed electronically via the Oncology Information Management System (MOSAIQ®) or on a designated cytotoxic order sheet, following a designated protocol and be overseen by a Consultant. All prescriptions must include the following⁴: ○ Age and / or date of birth ○ Current body weight and date weighed ○ Basis for the dose calculation e.g. mg/kg/dose (where appropriate) ○ Dose in unit of mass e.g. mg not mL Where they exist for more specialised purposes separate specific charts are to be used as approved by the Drug and Therapeutics Committee (DTC) e.g. Parenteral Fluid Therapy Order Chart, specialty insulin charts, specialty infusion charts, Inhalation Medication Sheet, Variable Rate IV Medication Infusion Chart, Agitation and Arousal Medication Chart, CHS414 Medication Administration Chart (for use in Education Support Services), Fluid Therapy Order Sheet – Cytotoxic Drugs, Downtime Cytotoxic Therapy Order Sheet,
Standardisation of Terminology, Abbreviations and Symbols in the Prescribing and Administration of Medicines Abbreviations for Clinical Documentation Policy Recommendations for terminology.	The use of inconsistent, ambiguous or non-standard abbreviations and terminology in the prescribing of medicines is a major cause of medication errors.⁴ Standardisation of terminology and abbreviations has been identified as an important strategy in reducing these errors.⁴

Systems	
System and Resources	Description
abbreviations and symbols used in medicines documentation.	
User applied Labelling of Injectable Medicines, Fluids and Lines Labelling of Injectable Medications and Fluids Policy	Labelling is a recognised risk in the safe administration of injectable medications. Preparation of injectable medications for bolus injections or infusion is complicated by multiple opportunities for error.⁴ Labelling of injectable medications is often not done or incomplete, omitting information such as name of medication, dose, patient name or time of preparation.⁴

Acknowledgements
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Related CAHS internal policies, procedures and guidelines
Allergy and Adverse Drug Reaction Documentation and Reporting (PCH Operational Manual)
Allergy and Adverse Drug Reaction Management (CAHS Policy Manual)
Antineoplastic (cytotoxic) Agents Safe Handling and Administration (PCH Medication Management Manual)
Children's Antimicrobial Management Program (ChAMP) (CAHS Policy Manual)
Clinical Incident Management (CAHS Policy Manual)
Diabetic ketoacidosis (PCH Clinical Practice Manual)
Guidelines for Drug Dosing in Overweight and Obese Children 2 to 18 Years of Age (PCH Medication Management Manual)
Guidelines for the use of the WA Clozapine Initiation and Titration Chart (WA Health)

Related CAHS internal policies, procedures and guidelines
Labelling of Injectable Medications and Fluids (PCH Medication Management Manual)
Medication Administration (PCH Medication Management Manual)
Medication Preparation, Checking and Administration (PCH Medication Management procedure).
Opioids, benzodiazepines and other Schedule 8 medicines (WA Health)
Pain Assessment (PCH Clinical Practice Manual)
Procurement, Distribution, Storage and Disposal of Medication (CAHS Policy Manual)
Psychotropic Medication - Monitoring Adverse Physical Health Effects (CAMHS Policy Manual)
Psychotropic Medication: Physical Health Monitoring Handbook CAMHS (CAMHS Medication Safety toolbox)
Schedule 8 and Restricted Schedule 4 Medication (PCH Medication Management Manual)

Medication Monographs (PCH Medication Management Manual)
Aciclovir Monograph - Paediatric
Adrenaline (epinephrine) Monograph – Paediatric
Alteplase Monograph - Paediatric
Amikacin Monograph - Paediatric
Aminophylline Monograph – Paediatric
Amiodarone Monograph - Paediatric
Atracurium Monograph – Paediatric
Blinatumomab Monograph - Paediatric
Calcium Monograph – Paediatric
Captopril Monograph - Paediatric
cARBOplatin Monograph - Paediatric
Cidofovir Monograph - Paediatric

Medication Monographs (PCH Medication Management Manual)
Cisplatin Monograph - Paediatric
Clobazam Monograph – Paediatric
Clonazepam Monograph - Paediatric
Clonidine Monograph – Paediatric
Clozapine Monograph – Paediatric
Colistimethate sodium Monograph - Paediatric
Colistimethate sodium (inhaled) Monograph - Paediatric
Cyclophosphamide Monograph for Rheumatology and Nephrology – Paediatric
Cytarabine Monograph - Paediatric
Dacarbazine Monograph - Paediatric
daCTINnomycin Monograph - Paediatric
Daunorubicin Monograph - Paediatric
Dexmedetomidine Monograph – Paediatric
Dexrazoxane Monograph - Paediatric
Diazepam Monograph – Paediatric
Digoxin Monograph – Paediatric
Dinutuximab Monograph - Paediatric
Dinutuximab BETA Monograph - Paediatric
Dobutamine Monograph – Paediatric
Dopamine Monograph – Paediatric
Doxorubicin Monograph - Paediatric
Droperidol Monograph - Paediatric
Enoxaparin Monograph – Paediatric
Etoposide Monograph - Paediatric
Etoposide PHOPHATE Monograph - Paediatric
Fentanyl Monograph – Paediatric
Flecainide Monograph - Paediatric
Ganciclovir Monograph - Paediatric

Medication Monographs (PCH Medication Management Manual)	
Gentamicin Monograph - Paediatric	
Heparin Monograph - Paediatric	
Hydromorphone Monograph - Paediatric	
Ifosfamide Monograph - Paediatric	
Insulin Monograph - Paediatric	
Irinotecan Monograph - Paediatric	
Isoprenaline Monograph – Paediatric	
Itraconazole Monograph – Paediatric	
Ketamine Monograph – Paediatric	
Levosimendan Monograph – Paediatric (requires SAS approval prior to prescription. See SAS online in PCH Pharmacy Hub)	
Liposomal amphotericin B (AmBisome) Monograph - Paediatric	
Liposomal amphotericin B (inhaled) Monograph (AmBisome) - Paediatric	
Lithium Monograph – Paediatric	
Magnesium Monograph – Paediatric	
Metaraminol Monograph – Paediatric	
Midazolam Monograph – Paediatric	
Milrinone Monograph – Paediatric	
Morphine Monograph – Paediatric	
Natalizumab Monograph - Paediatric	
Nivolumab Monograph - Paediatric	
Noradrenaline (norepinephrine) Monograph - Paediatric	
Oxycodone Monograph – Paediatric	
Pegaspargase Monograph - Paediatric	
Pemetrexed Monograph - Paediatric	
Pethidine (intravesical) Monograph – Paediatric	
Phenobarbital (phenobarbitone) Monograph – Paediatric	
Phenytoin Monograph - Paediatric	
Phosphate Monograph – Paediatric	

Medication Monographs (PCH Medication Management Manual)
Posaconazole Monograph - Paediatric
Potassium Chloride Monograph – Paediatric
Propofol Monograph – Paediatric
Rifampicin Monograph - Paediatric
Rituximab Monograph - Paediatric
Rocuronium Monograph – Paediatric
Sodium chloride 3% and 23.4% (Hypertonic Saline) Monograph - Paediatric
Suxamethonium Monograph – Paediatric
Taurolidine/Sodium Citrate/Heparin 100 units (TaurolockHep100®) Monograph - Paediatric
Teicoplanin Monograph – Paediatric
Tobramycin Monograph – Paediatric
Tobramycin (inhaled) Monograph - Paediatric
Valaciclovir Monograph – Paediatric
Valganciclovir Monograph – Paediatric
Vancomycin (intravenous) Monograph - Paediatric
Vancomycin (inhaled) Monograph - Paediatric
Vancomycin (oral) Monograph - Paediatric
Vecuronium Monograph – Paediatric
vinBLASTine Monograph – Paediatric
vinCRISTine Monograph – Paediatric
vinORELBine Monograph – Paediatric
Voriconazole Monograph - Paediatric
Warfarin Monograph - Paediatric
Zuclopenthixol Monograph - Paediatric

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Useful resources (including related forms)
Australian Commission on Safety and Quality in Health Care position statement on paediatric prescribing
Gentamicin and Tobramycin Reporting Form MR860.91
Medication Chart Policy (effective from 31 March 2023) MP 0078/18 (WA Department of Health)
Medication Management and Pharmacy (PCH HealthPoint Information Hub)
Mandatory Standard for Intravenous Potassium (WA Department of Health)
Mandatory Standard for Vinca Alkaloids (WA Department of Health)
Opioids, benzodiazepines and other Schedule 8 medicines (WA Department of Health)
Specialty Infusion Charts (PCH Pharmacy HealthPoint Information Hub)

This document can be made available in alternative formats on request.

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